**ANSWER KEY**

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · W1 · WORKING SCIENTIFICALLY

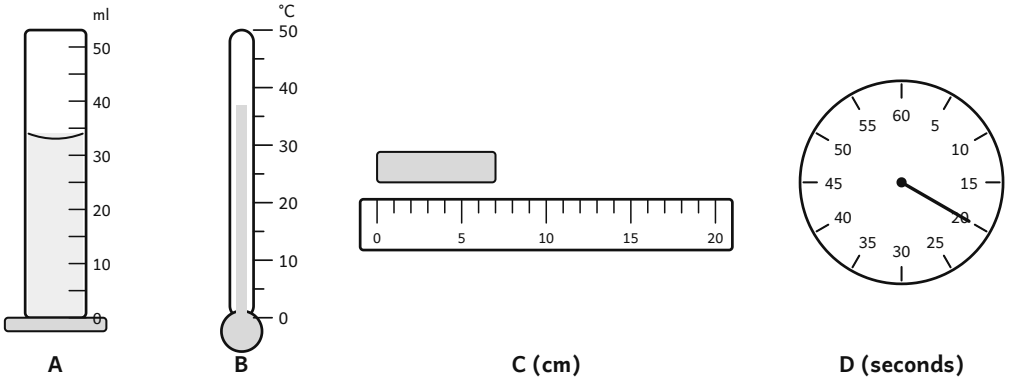
# Tools, units and tests

★☆☆
Grade 3

**Learning goals**

- Match tools and units.
- Spot testable questions.

## 1 Read each instrument. Include the unit. [4]




A B C (cm) D (seconds)

| A | B | C | D |
|---|---|---|---|
| 34 ml | 37 °C | 7 cm | 20 s |

## 2 Draw a line to match the tool to the unit. [4]

Draw a line to match each pair, or write the letter.

| | | |
|---|---|---|
| thermometer ● | → A | ● **A** °C |
| stopwatch ● | → C | ● **B** cm |
| ruler ● | → B | ● **C** seconds |
| forcemeter ● | → D | ● **D** newtons |

## 3 Can it be tested? Sort the questions. [4]

SORT THESE

**Which ball bounces highest?** **Does salt melt ice faster?**
**Is red better than blue?** **Which song is the nicest?**

| Can be tested | Cannot be tested |
|---|---|
| Which ball bounces highest? | Which song is the nicest? |
| Does salt melt ice faster? | Is red better than blue? |

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · W1 · WORKING SCIENTIFICALLY

# Making it fair

★★☆
Grade 3

**Learning goals**

- Plan a fair test.
- Explain repeats.

**1** **Lina wants to know which cup keeps cha hot longest. She puts hot cha in a clay cup and cold cha in a steel cup. Is it fair? How can she fix it?** [3]

No: the starting temperature is different. She must start both with the same amount of cha at the same temperature.

**2** **Why should you repeat a measurement three times?** [2]

To check the results and make them more reliable.

**3** **Write a prediction with a reason: "Will a plant grow taller in sunlight or in shade?"** [2]

I think it will grow taller in sunlight because plants need light to make food.

ANSWER KEY

INVESTIGATE · W1 · WORKING SCIENTIFICALLY

# Which paper makes the best paper plane?

★★☆ Grade 3

**Learning goals**

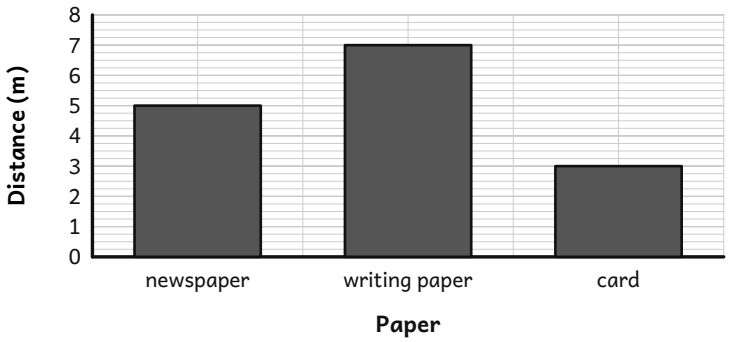
- Repeat measurements.
- Draw a bar chart.
- Say what the evidence shows.

Class 3A made the same paper plane from three kinds of paper and threw each one three times.

**1** **Complete the table. Find the middle (median) result of each.** [3]

| Paper | Try 1 (m) | Try 2 (m) | Try 3 (m) | Middle result (m) |
|---|---|---|---|---|
| newspaper | 4 | 5 | 6 | 5 |
| writing paper | 6 | 8 | 7 | 7 |
| card | 3 | 2 | 4 | 3 |

**2** **Draw a bar chart of the middle results.** [3]




**3** **Which paper was best? What does the evidence show?** [2]

Writing paper flew furthest (7 m), so it makes the best plane.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · W1 · WORKING SCIENTIFICALLY

# Testing umbrellas for the monsoon

★★★
Grade 3

**Learning goals**

- Plan a real test.
- Use evidence to choose.

Science | Design | Maths

The monsoon rains are coming. A shopkeeper in Rajshahi wants to sell the best umbrellas. He asks a Grade 3 class to test three kinds of cloth: cotton, plastic sheet and nylon.

The class drips 50 drops of water onto each cloth stretched over a cup, then measures how much water gets through.

A story about a class investigation. Results are example data.

**1** **Complete the plan.** [3]

| | |
|---|---|
| Change | the type of cloth |
| Measure | water that gets through (ml) |
| Keep the same | number of drops, size of cloth, cup |

**2** **Results: cotton 8 ml, plastic 0 ml, nylon 1 ml. Which cloth should the shopkeeper use? Why?** [2]

Plastic sheet (or nylon): almost no water gets through.

**3** **What else should they test before choosing?** [1]

Strength, weight, cost, whether it tears in wind.

**ANSWER KEY**

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · W1 · WORKING SCIENTIFICALLY

# Show what you know

★★★
Grade 3

**Learning goals**

- Use science words.

## 1 Find the words.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| K | E | K | R | E | P | E | A | T |
| F | T | E | B | Y | J | A | I | C |
| A | H | K | I | W | F | B | T | H |
| I | G | P | T | E | S | T | A | A |
| R | D | P | B | R | O | R | B | R |
| P | R | E | D | I | C | T | L | T |
| F | J | P | G | G | D | Y | E | T |
| H | M | E | A | S | U | R | E | J |
| U | T | H | M | W | A | L | W | S |

**chart fair measure predict repeat table test**

## 2 Which tool measures time? [1]

○ thermometer
● **stopwatch**
○ ruler

## 3 In a fair test you change... [1]

● **one thing**
○ two things
○ everything

## 4 Challenge. Plan a test: "Which ball rolls furthest?" Write what you change, measure and keep the same. [3]

| | |
|---|---|
| Change | the type of ball |
| Measure | distance rolled (cm) |
| Keep the same | ramp, push (none), floor |